

ORIS EMR

Next-Generation Dental Chamber Management & Clinical Operating System purpose-built for Bangladesh Dental Practices.

✓ Sub-60s Bangla Prescriptions

✓ Barcode Scanning ID

✓ Smart Slot Anti-Collision

✓ Live Waiting TV Lounge

✓ Whole-BDT Dental Billing

TARGET AUDIENCE

**Dental Surgeons, Clinic
Owners & Content
Creators**

DOCUMENT VERSION

2.0 Production Ready

VIDEO PRODUCTION LENGTH

Suggested: 5 to 7 Minutes

1. Platform Architecture & Role Hierarchy

FOUNDATION

ORIS EMR is structured as a high-security, multi-tenant cloud application. Each dental chamber operates in strict tenant isolation, with its own operational schedules, doctor rosters, patient records, catalog procedures, and letterhead configurations.

**Superadmin (Platform Owner)**

Manages chamber onboardings at /platform.
Controls clinic subscription tiers, feature flags, global doctor catalogs, and tenant suspension status.

**Tenant Admin (Chamber Owner / Lead Dentist)**

Full ownership of the chamber: configs clinic logo, working hours, doctor personal schedules, physical chair count, letterhead margins, and staff user logins.

**Clinical Dentist (Doctor)**

Access to in-chair queue, patient history, FDI tooth chart, Bangla prescription builder, in-chair radiograph Lightbox, and confidential doctor's private clinical notes.

**Front Desk / Receptionist**

Rapid barcode card scanning, new patient registration, appointment booking, chair queue management, bill settlement, payment collection, and dues recording.

Unique Regional Adaptations for Bangladesh Dental Chambers

CAPABILITY	STANDARD GLOBAL EMR	ORIS EMR BANGLADESH SPECIALIZATION
Prescription Printout	Generic English names & instructions	Localized Bangla instruction bullets (e.g. ি+০+ি (ভরা পেটে)) with printed BMDC Registration.
Payment Currencies	Cents/Decimals, credit card only	Whole BDT (৳) rounded amounts with native support for bKash, Nagad, Rocket, Cash.
Patient Card Numbers	Random UUID / arbitrary numbers	Pre-printed batch barcode cards (scannable via 1D/2D reader) or collision-proof auto-increment cards.
Phone Validation	International E.164 only	Native 11-digit Bangladeshi mobile format (01XXXXXXXX) with automatic sanitization.

2. The 5-Phase End-to-End Clinic Workflow

CORE JOURNEY

Content creators should anchor their video around this smooth, friction-free journey showing a patient from arrival to billing.

1

Patient Check-In & Barcode Identification (Reception)

Patient enters chamber. Receptionist scans the physical chamber card barcode (or searches by mobile number). System instantly opens the record, confirms active appointment, and places them into the **Live Waiting Queue**.

2

Live TV Waiting Lounge & Audio Chime

Lobby TV screen displays the live queue with estimated waiting times. When the dentist clicks *"Call to Chair"*, a pleasant audio chime sounds, guiding the patient to their designated Dental Chair (Chair 1, Chair 2, etc.).

3

In-Chair Consultation & Radiograph Lightbox (Doctor Workspace)

Dentist accesses patient dossier right from the dental chair. Inspects previous X-rays and OPGs via the built-in Lightbox (with 360° rotation and zoom). Checks drug allergies (Penicillin, Sulfa) flagged prominently in red.

4

Sub-60-Second Bangla Prescription Issuance

Doctor clicks affected teeth on the **FDI Two-Digit Dental Arch**, selects diagnosis via typeahead autocomplete, picks catalog medicines with Bangla dosage templates, writes confidential internal notes, and prints the official prescription.

5

Automated Settlement & Billing (Whole-BDT)

Patient automatically moves from in-chair to **Billing**. Line items for performed procedures are auto-populated. Cashier settles full or partial balance (Cash/bKash), records previous dues, and prints a branded receipt with barcode.

3. Clinical Prescription Builder Deep Dive

DENTIST FAVORITE

🦷 FDI Dental Arch (32-Tooth Selector)

- Upper Right (11-18), Upper Left (21-28)
- Lower Left (31-38), Lower Right (41-48)
- Multi-select teeth with visual highlighted badges
- Prints explicitly on patient prescription paper

⚡ Typeahead Clinical Autocomplete

- **C/C:** Toothache, Cavity, Bleeding gum
- **O/E:** Deep caries, Percussion positive
- **Diagnosis:** Irreversible pulpitis, Gingivitis
- **Investigations:** IOPA X-Ray, OPG Panoramic

🔒 Doctor's Private Clinical Notes (Confidential)

A high-demand feature added specifically for dentists. Doctors can type internal thoughts, diagnostic suspicions, or procedural notes that are **strictly stored in the EMR**.

Security Guarantee: This note is visible in the internal EMR via the **"View"** modal, but is *never* printed onto the patient's paper prescription or receipt.

Drug Allergy Safety Radar

When prescribing, the system cross-references patient allergies (e.g. Penicillin, NSAIDs) with the active ingredient and drug class of catalog medicines in real-time.

⚠️ **Warning Notice:** Displays an alert banner when prescribing a medication with mild or moderate potential cross-reactivity.

🔴 **Hard Allergy Block:** Halts prescription submission if prescribing a known severe allergen, requiring intentional override logging.

Bangla Advice & Post-Op Guidelines

Pre-configured standard advice blocks for extraction, RCT, scaling, and orthodontic care that paste into the prescription in 1 click (e.g., *তুলা ১ ঘণ্টা শক্ত করে চেপে রাখবেন, গরম বা শক্ত খাবার খাবেন না*).

4. Precision Scheduling & Online Booking

ANTI-COLLISION

ORIS EMR replaces manual appointment notebooks with a mathematical slot calculation engine that prevents double-booking and wasted chair time.

Dynamic Slot Engine Algorithm

- **Split Shifts Support:** Morning Shift (10:00 AM - 02:00 PM) & Evening Shift (05:00 PM - 09:00 PM). Automatically blocks lunch/break windows.
- **Dynamic Duration by Procedure:** Tooth Extraction (30 mins), Scaling (45 mins), Consultation (15 mins). Timeslots adapt based on selected procedure combo.
- **Buffer Spaces:** Configurable 5-15 minute sterilization buffer between consecutive appointments.
- **Real-time Clock Check:** When booking for today, slots earlier than current time are automatically removed.

Public Booking Portal (/book/[tenantSlug])

New Patient Online Booking

Patients booking online enter Name and Phone. The system **automatically registers them into the EMR** with a collision-proof digital card sequence. They show up across all appointment lists and queues with full name and patient card!

Existing Cardholders Auto-Fill

Existing patients simply enter their **Chamber Card ID**. The system instantly verifies their record, auto-populates their full name and mobile phone, and confirms their booking in seconds.

Public Waiting Lounge (/lounge/[tenantSlug]): A dedicated URL designed to run fullscreen on a TV in the clinic waiting area, showing currently serving token numbers, patient initials, and upcoming queue.

5. Video Creator Script & Production Blueprint

YOUTUBE / PROMO READY

Use this timestamped sequence for recording a high-converting software review or tutorial video.

[00:00 - 00:45] INTRO & THE PROBLEM

Visual: Fast-paced B-roll of paper prescriptions, cluttered registers, waiting room confusion.

Voiceover: "Running a dental chamber in Bangladesh shouldn't mean drowning in paper notes, conflicting appointments, and lost medical histories. Today we're reviewing ORIS EMR – the dental operating system built specifically for modern dental clinics in Bangladesh."

[00:45 - 01:45] BARCODE CHECK-IN & LIVE QUEUE

Visual: Screen capture of scanning patient card barcode. Live queue updates instantly.

Voiceover: "Watch this: A patient walks in. With one beep of the barcode scanner, their profile is pulled up. Reception assigns them to Dental Chair 1. Over in the waiting lobby, the TV lounge screen updates with an audio chime!"

[01:45 - 03:30] CLINICAL DENTIST WORKSPACE & PRESCRIPTION

Visual: Dentist screen: opening in-chair X-ray lightbox, clicking teeth on FDI chart, typeahead suggestions.

Voiceover: "Now the doctor's perspective: X-rays can be inspected right in chair. The interactive FDI arch lets you click affected teeth. Medicine dosages appear in clear Bangla. And look at this – confidential private notes the doctor can save that never print on the patient's paper."

[03:30 - 04:45] DENTAL BILLING & BDT SETTLEMENT

Visual: Move patient to billing. Procedures auto-populate. Record bKash payment.

Voiceover: "As soon as the prescription is saved, the patient moves to billing. Notice how the billable procedures are already loaded. You can collect full payment or record outstanding dues in whole Taka. Professional printout with barcodes in one click."

[04:45 - 06:00] ONLINE BOOKING & SUMMARY

Visual: Mobile view of public booking page. Demonstrating anti-collision slots.

Voiceover: "Patients can also book appointments online from home. The smart slot engine handles split shifts so double bookings are literally impossible. ORIS EMR is lightning-fast, paperless, and purpose-built."

6. Technical Summary & Branding Assets

REFERENCE

Where to Place Clinic & Application Logos

LOGO DESTINATION	SUPPORTED FORMATS	FILE LOCATION / UPLOAD PATH
Global Application / Platform Branding (Landing Page, Login, App Bar)	.png, .svg	Keep file in /public/logo.png or /public/logo.svg. Next.js automatically serves this at /logo.png.
Chamber Print Letterhead & Email Logo (Prescriptions, Invoices, Patient Emails)	.jpg, .png, .webp, .svg (Max 5MB)	Upload directly in web app: Chamber Settings (/app/settings) → Clinic Logo Upload . Stored securely in MinIO/S3 with instant print preview.

Technology Stack Highlights

Frontend

Next.js 15 (App Router),
TypeScript, Tailwind CSS, Framer
Motion animations.

Database & ORM

PostgreSQL (Supabase/Neon),
Drizzle ORM, GIST exclusion
constraints.

Storage & Barcodes

S3 / MinIO Cloud Storage for X-
rays, JsBarcode Code-128 SVG
generator.



Contact & Support

For technical assistance, chamber onboarding, or feature inquiries, reach out to the ORIS EMR Development Team.